

1. Definition and characteristics of cloud computing

cloud computing is the delivery of computing resources such as servers, storage, databases, networking software, and processing power over the internet.

cloud computing differs from traditional on-premise computing through the following characteristics.

- on-demand self-service:

users can create or access computing resources such as virtual machines and storage without directly contacting the service provider.

- Broad network access:

cloud services are available through the internet using computers, mobile phones, tablets and other devices.

- Resource pooling:

the provider shares a large pool of computing resource among multiple users using virtualization.

- Rapid elasticity:

cloud resources can be quickly increased or decreased according to workload requirements.

2. Evolution of cloud computing

cloud computing evolved from main frames to personal computers, computer networks, virtualization and the Internet.

These advances led to modern cloud platforms like AWS, Microsoft Azure, and Google Cloud.

Paas (Platform as a service)

provides a ready-made platform for developers to build, test and deploy Applications without managing the underlying hardware and operating system

Example: Google App Engine.

SaaS (software as a service)

The growth of the world wide web and Internet based software enabled applications to be accessed through web browsers.

Thus, the evolution from mainframes to pc's networks, virtualization, high-speed internet and web based software created the foundation for modern cloud computing.

3. Role of server virtualization in cloud computing
server virtualization divides one physical server into multiple virtual machines using a hypervisor. It improves hardware use, flexibility, scalability and reduces cost.

However virtualization builds the infrastructure, while cloud computing delivers it as a service.

Virtualization and cloud computing are not the same. virtualization is an enabling technology that creates virtual computing.

cloud computing is a service delivery model that provides computing resources over a network, usually with on-demand access, elasticity, resource pooling, and pay-per-use pricing.

Therefore, virtualization helps build the infrastructure while cloud computing provides that infrastructure as a service to user.

4. Role of Data centers in delivering cloud services

Data centers contains resources, storage, networking, cooling and back system. They provide secure, reliable and scalable cloud services with high availability and disaster recovery.

They also provide scalability, allowing resources to be increased with demand grows. Multiple data centers located in different geographical regions improve availability and reduce service interruptions.

Important features of cloud data centers include high-speed networking, backup power, cooling systems, security controls, data backup, and disaster recovery mechanisms.

Thus, data centers act as the physical infrastructure that supports the reliable, secure and continuous delivery of cloud services.

5.

Difference between IaaS, PaaS, SaaS and XaaS

cloud services are commonly divided into different service models.

IaaS (Infrastructure as a service)

provides basic computing infrastructure such as virtual machines, storage, and networking.

Example: Amazon EC2.

PaaS (Platform as a services):

provides a ready-made platform for developers to build, test and deploy applications without managing the underlying hardware and operating system.

Example: Google App Engine.

SaaS (software as a service)

provides complete software applications through the internet. users simply access the application while the provider manages the software and infrastructure.

Example: Google workspace or microsoft 365.

XaaS (Everything as a service):

A broad concept in which almost any IT resource or technology can be delivered as an online service. Examples include database as a service, storage as a service, security as a service and artificial intelligence as a service.

Example: Amazon RDS as database as a service.